**PROBLEM STATEMENT #01 | Campus & Academic Operations**

# Smart Lab Equipment & Slot Reservation Portal

## 1. Problem Context & Overview

Engineering lab facilities require a specialized scheduling mechanism to prevent double-booking of high-value hardware instruments (oscilloscopes, logic analyzers, and FPGA development boards). The system must handle slot reservations, verify equipment calibration status, and enforce late-return disciplinary rules.

**Target Stakeholders / Actors:** *Student, Lab Technician*

## 2. Sample Functional Requirement (FR) Guideline

**Sample Requirement: FR-001 [Priority: High]**
Description: The system shall allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance.
Sample Acceptance Criteria: Pass: Slot is successfully locked and reservation token generated; Fail: Double-booking allowed or reservation exceeds 2-hour limit.

## 3. Sample Non-Functional Requirement (NFR) Guideline

**Sample Requirement: NFR-001 [Type: Performance & Security]**
Description: The system shall ensure that concurrent slot reservation requests are processed within 200 ms with strict database-level lock isolation to prevent race conditions.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

**Deliverables to Submit: [All through a GitHub Repo in your account]**

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.